

M003 Extension 2 — English Worksheet (Advanced)

Topic: 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Advanced (Extension, after Week 3) · **Time:** about 45 minutes

Build **two mob heads** in 3D — a **creeper** and a **piglin**. Every block needs **three** numbers: **(x, y, z)**. x is **across**, y is **up**, z is **deeper** (forward). Both heads stand in front of you in the world. Walk around them, then copy them block by block. You build the **front face** first, then repeat it at z = 2 and z = 3 to add depth — the face shows on the front only.

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🧠

Read each set. Do **not** run it. Write your prediction **and the reason**.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Does the line go across, up, or deeper? Why?


```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
place green block at (4, 7, 1)
```

Across, up, or deeper? Why?


```
place green block at (3, 5, 1)
place green block at (4, 5, 1)
place green block at (3, 5, 2)
place green block at (4, 5, 2)
```

How many blocks, and what shape do they make? Why?


```
place green block at (3, 5, 1)
place black block at (3, 5, 1)
```

Same (x, y, z). Green, black, or both? Why?

2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

Bug A — This block should sit one step **deeper**, at (3, 5, 2). It only has two numbers, so it lands on the front.

```
place green block at (3, 5)
```

Fixed code:

Why wrong? Why does your fix work? (2–3 sentences)

Bug B — The eye should be **up high and one step deep**: (4, 7, 2). The last two numbers are swapped.

```
place black block at (4, 2, 7)
```

Fixed code:

Why wrong? Why does your fix work? (2–3 sentences)

Bug C — The code runs with no error, but the creeper's eye ended up on the **side** of the head instead of the front. The front face is built at $z = 1$, but this eye was placed at $z = 2$ — one block too forward.

```
place green block at (4, 7, 1)
place black block at (4, 7, 2)
```

Fix it so the eye is on the front face, and explain what "one block too forward" means. (2–3 sentences)

3 · Build It 📷

Build **both** heads on your home spot — a **Creeper** and a **piglin** — then add one change of your own.

Plan your coordinates first. Pick a home block for each head so they don't overlap.

Creeper: build the front face at $z = 1$ (green square + black eyes + black mouth), then repeat at $z = 2$ and $z = 3$. Plain green where the face was.

Piglin: build the front face at $z = 1$ (pink square + a wide black mouth + nose), then repeat at $z = 2$ and $z = 3$. Plain pink where the face was.

Face on front only: at $z = 2$ and $z = 3$, fill the face spots with the plain head colour.

Then a MODIFY challenge: change one thing your own way — a new colour, an extra part (ears, hat), or make a head bigger.

💡 Tip: the eyes, nose, and mouth only show on the **front**. The sides and back are plain.

Send a photo or video. Then finish these 5–6 sentence starters.

I built the ... and the ...

The lowest front block of my creeper is at (x = ..., y = ..., z = ...).

To go up I changed the ... number; to go deeper I changed the ... number.

Each head is ... blocks deep, and the face is on the ... side only.

For my MODIFY challenge I ...

The hardest part was ... because ...

I built the ... and the ...

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For my MODIFY challenge I ...

The hardest part was ... because ...

4 · Explain It 🎥

Film yourself showing both heads. Use these words: **x, y, z, across, up, deeper, front face, home spot.**

1. Show your home spot. Say why you stand on it every time.
2. Walk around each head — show the front, a side, and the back.
3. Point at one block: "this one is at x ..., y ..., z ...".
4. Say in your own words why the face is on the front only, and show your MODIFY change.

Write a short paragraph (5–6 sentences) about how you built your heads:

If you taught a friend to build a 3D head, what would you teach first, and why?

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.